

Dhruvil Satasiya

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OBJECTIVE

Machine Learning Research Engineer focused on transformer interpretability, adversarial evaluation, and AI safety. Experienced in implementing transformer architectures from scratch, conducting mechanistic interpretability experiments, reproducing frontier ML research, and translating research prototypes into production ML systems. Interested in robust model evaluation, representation learning, and trustworthy deployment of large language models.

EDUCATION

•Indian Institute of Technology Madras (IITM), Chennai

2023–2026

BS in Data Science

Relevant Coursework: Machine Learning Practice, Statistics, Data Structures, DBMS, Python Programming

•P P Savani University, Surat

2022–2026

B.Tech in Information Technology

Relevant Coursework: Artificial Intelligence, Optimization Techniques, Big Data Analytics

INTERNSHIP EXPERIENCE

•AI/ML Intern

June 2025–Nov 2025

On-site

10Turtle

- Designed and deployed production LLM systems with improved latency, cost efficiency, and response quality.
- Ran controlled experiments across model configurations to optimize performance in real-world settings.
- Built containerized inference services with automated logging, health checks, and monitoring to ensure >99% uptime.
- Trained and deployed YOLO-based object detection models for multi-class image analytics, optimizing data augmentation and inference pipelines.
- Converted experimental LLM workflows into production-ready systems using reproducible evaluation and logging pipelines.

•SDE Intern

Jan 2024–may 2025

Remote

Ashirwad Infotech

- Deployed a fine-tuned LLM and an open-source image generation model on GPU instances using vLLM, exposing inference APIs for low-latency serving.
- Optimized inference by resolving GPU OOM issues and evaluating 4-bit/8-bit quantization and dynamic batching configurations to improve throughput and memory efficiency.
- Fine-tuned an instruction-following LLM using SFT with LoRA/QLoRA for parameter-efficient adaptation.
- Prepared chat-style instruction datasets, including formatting, tokenization, and prompt template construction using Hugging Face and PyTorch.
- Managed the complete fine-tuning workflow from data preparation to evaluation and adapter merging.

PERSONAL PROJECTS

•Mechanistic Interpretability of Transformer with Multi-Head Latent Attention (MLA)

Source Code

- Built and trained a 114M-parameter GPT-style decoder transformer from scratch in PyTorch using modern LLM components including RoPE, SwiGLU, and RMSNorm.
- Engineered a custom Multi-Head Latent Attention (MLA) mechanism with low-rank KV compression, reducing inference memory usage by 81% while building optimized training pipelines and done SFT with TinyStories with efficient CUDA utilization and stable scaling via muP.

•Activation Probe Safety Monitor

Source Code

- Built a lightweight LLM safety monitor using linear probes trained on hidden activations from harmful and benign instruction datasets.
- Collected intermediate LLM activations using PyTorch forward hooks and trained layer-wise Logistic Regression probes with scikit-learn.
- Deployed the best-performing probe as a FastAPI sidecar for real-time inference-time safety classification without modifying the underlying LLM.

•**Greedy Coordinate Gradient (GCG) Implementation**

Source Code

- Implemented the Greedy Coordinate Gradient (GCG) adversarial attack in PyTorch to generate suffix-based jail-break prompts for instruction-tuned LLMs.
- Built a FastAPI service exposing the attack pipeline and evaluated attack success rates on a subset of AdvBench harmful behaviors.

•**PAIR Adversarial Evaluation Framework**

Private repository; withheld for responsible disclosure

- Implemented the PAIR algorithm for black-box LLM adversarial testing using an iterative multi-agent (attacker–judge–target) framework
- Deployed a FastAPI + Docker microservice achieving a 72% attack success rate on AdvBench with an average of 11.4 iterations.

TECHNICAL SKILLS

- **Programming:** Python, SQL
- **Libraries/Frameworks:** PyTorch, Transformers, Accelerate, PEFT, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
- **NLP & LLM Tools:** LlamaIndex, LangChain, Prompt Engineering, OpenAI APIs
- **ML Techniques:** Model Fine-tuning, A/B Testing, Object Detection, Sequence Modeling, Experiment Design
- **Deployment:** Docker, Logging/Monitoring, API Integration, Git
- **Soft Skills:** Clear technical communication, research mindset, rapid prototyping, teamwork

CERTIFICATIONS

- Reinforcement Learning** NPTEL (IIT Madras) | Elite Certification *Jan-Apr 2025*
- AWS Academy Cloud Foundations** AWS Academy *Mar 2025*